

Programme	B. Sc. Mathematics Honours			
Course Code	MAT2MN105			
Course Title	VECTOR SPACES AND LINEAR TRANSFORMATIONS			
Type of Course	Minor			
Semester	II			
Academic Level	100 – 199			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Linear Algebra Course in Semester 1 - Vectors and Matrices			
Course Summary	This course delves into advanced concepts in linear algebra, focusing on general vector spaces, basis and dimension, matrix transformations, and eigenvalues and diagonalization. The course builds on foundational linear algebra principles and explores their applications in higher-dimensional spaces and complex transformations.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and apply concepts related to vector spaces, including understanding vector space axioms, subspaces, and the solution space of homogeneous systems.	U	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO2	Explore the concepts of linear independence, coordinates, basis, and dimension within vector spaces, including computing basis vectors and understanding coordinate systems relative to a basis.	Ap	P	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO3	Analyse and apply matrix transformations, including basic transformations in R^2 and R^3 , understanding properties of these transformations, and exploring concepts related to eigenvalues, eigenvectors, and diagonalization of Matrices.	An	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)# - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text: Howard Anton and Chriss Rorres, Elementary Linear Algebra (11/e), Applications version, Wiley				
Module	Unit	Content	Hrs (60)	Ext. Marks (70)
I	General Vector Spaces		12	
	1	Section 4.1: -Real vector spaces – up to and including Example 8.		
	2	Section 4.1:- Rest of the section.		
	3	Section 4.2: - Subspaces (examples 7, 8 are optional) – up to and Example 10.		
	4	Section 4.2: - From Example 10 to Example 15 (proof of theorem .4.2.3 is optional)		
	5	Section 4.2: - Rest of the section (Linear transformation view point is optional)		
II	Basis And Dimension		12	
	6	Section 4.3: - Linear independence – up to and including Theorem 4.3.3		
	7	Section 4.3: - Rest of the section (proofs of all the results are optional).		
	8	Section 4.4:- Coordinates and Basis -up to and including Example 5		
	9	Section 4.4: - rest of the section from Theorem 4.4.1.		
	10	Section 4.5:-Dimension – up to and including Example 3.		
	11	Section 4.5: - Rest of the section from Example 3 (proofs of all the theorems are optional).		
III	Matrix Transformations		12	
	12	Section 4.9: - Basic matrix transformations in R^2 and R^3 - Reflection operators, Projection operators		
	13	Section 4.9:- Rotation Operators – Rotation in R^3		
	14	Section 4.9:- Rest of the section.		
	15	Section 4.10: - Properties of Matrix Transformations – up to and including Example 4.		
	16	Section 4.10:- rest of the section (proofs of theorems are optional)		
	17	Section 4.11: - Geometry of Matrix Operators on R^2 (proof of Theorem 4.11.2 is optional)		
IV	Eigen Values and Diagonalization		12	
	18	Section 5.1:- Eigen values and eigen vectors – up to Theorem 5.1.3		
	19	Section 5.1; -From Theorem 5.1.3 to Example 7 (including)		
	20	Section 5.1: - Rest of the section (Eigen values of general linear transformation is optional)		
	21	Section 5.2: - Diagonalization – up to and including Example 4 (proofs of theorems are optional)		
	22	Section 5.2; - Rest of the section (Geometric and algebraic multiplicity are optional)		
V	OPEN ENDED		12	
	Rank space, Null space and Rank- Nullity theorem, General Linear transformations and Matrix representation, Eigen values of general linear transformation, Geometric and algebraic multiplicity.			

References:

- 1 Advanced Engineering Mathematics, 6th Edition, Dennis G. Zill, Jones & Bartlett Learning LLC (2018) ISBN: 978-1-284-10590-2.
2. Advanced Engineering Mathematics, Erwin Kreyzsig, 10th Edition, Wiley India.
3. Linear Algebra and its Applications: 3rd Edition, David C. Lay, Pearson Publications

Note: 1) Optional topics are exempted for end semester examination. (2) Proofs of all the results are exempted for external exam. (3) 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	3	1	1	1	3	0	0
CO 2	2	1	2	1	1	1	2	0	0
CO 3	2	1	3	1	1	1	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓